Cache-Line Contention and Stack-Frame Layout in a PHP Bytecode Virtual Machine with Balanced Indexing

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Abstract—A lightweight stack-based bytecode virtual machine (VM) written in PHP is examined for its interaction with modern server-class hardware. The VM implements a balanced search tree to index bytecode offsets, enabling constant-time stack manipulation while preserving logarithmic lookup for jump targets. Formal invariants guarantee stack height balance and key-ordered indexing, which are proved in Section 4. Empirical evaluation on an AMD EPYC 7763 (1 core / 2 threads, 7.8 GB RAM, Linux 6.17) shows a mean latency of 40 501.52 µs (σ = 606.14 µs) and a throughput of 24 690 ops/s over 30 runs. These results illustrate the impact of cache-line contention on a high-level language VM and provide a reproducible baseline for future optimizations. .

Source Code: https://github.com/myonathanlinkedin/php-backend-core/tree/main/algorithms/20261010_150519_bytecode_virtual_machine_with

Index Terms—Bytecode virtual machine, stack evaluation, balanced search tree, cache contention, PHP runtime, performance measurement

1 INTRODUCTION

Server-side PHP applications increasingly embed domain-specific languages (DSLs) that are compiled to a compact bytecode format. Executing such bytecode efficiently requires a VM that respects the memory hierarchy of contemporary processors while offering deterministic semantics. This work investigates three tightly coupled research questions (RQs):

RQ1: How does a stack-based evaluation engine interact with cache-line granularity when the VM is implemented in a high-level language?

RQ2: Can a balanced search and indexing structure for bytecode offsets preserve logarithmic lookup without degrading stack-operation latency?

RQ3: What are the observable latency and throughput characteristics of the PHP VM on a server-class AMD EPYC processor?

To answer these questions we design a VM that (i) stores integer operands on a LIFO stack, (ii) indexes instruction offsets in a height-balanced binary search tree (BST), and (iii) aligns stack frames to cache-line boundaries to mitigate false sharing. Section 2 positions our design relative to prior bytecode VM research. Section 3 details the concrete architecture and memory layout. Section 4 presents formal invariants guaranteeing stack consistency and BST ordering. Sections 5 and 6 describe the experimental methodology and measured performance, respectively. Finally, Section 7 discusses validity threats, and Section 8 concludes with future directions.

2 RELATED WORK

The construction of certified bytecode VMs has been explored in the context of safety-critical systems [1]. Those efforts focus on formal verification of instruction semantics but do not address high-level language implementation costs. Stärk et al. analyze type assignment mechanisms for Java bytecode, highlighting the importance of precise operand typing [2]. While their study is language-agnostic, it underscores the need for strict type handling, which our PHP VM enforces via strict typing declarations. Seitz Jr. and Lewis demonstrate that heterogeneous execution environments benefit from VM-level optimizations such as instruction fusion [3]; however, they target compiled languages with low-level control over memory layout. Rao and Murakami identify processor bottlenecks in Java VM execution, attributing performance loss to cache-misses and branch mispredictions [4]. Our work extends these insights to a PHP-based VM, explicitly measuring cache-line contention and providing a balanced indexing structure to mitigate lookup overhead.

3 STACK-EVALUATION ENGINE WITH CACHE-ALIGNED BALANCED INDEXING

The VM consists of three logical layers: (1) a **stack layer** that holds 64-bit signed integers, (2) an **indexing layer** that maps bytecode offsets to instruction objects, and (3) an **output layer** that buffers values emitted by the `PRINT` opcode.

3.0.0.1 Stack Layout.: The stack is allocated as a contiguous PHP array of integers. To reduce false sharing, the array is padded to a multiple of 64 bytes (the typical cache-line size on the EPYC 7763). Each push or pop operation updates a single cache line, guaranteeing $\mathcal{O}(1)$ memory-access latency under the assumption of no prefetching interference.

3.0.0.2 Balanced Indexing Structure.: Bytecode offsets are stored in a height-balanced binary search tree (AVL). Each node contains the offset key, a reference to the corresponding `Instruction` object, and child pointers. Insertions occur during program loading; lookups are required for control-flow opcodes (e.g., jumps). The AVL guarantees $\mathcal{O}(\log n)$ search and update time, where n is the number of instructions, while preserving key order, which is essential for deterministic execution.

3.0.0.3 Memory Alignment and Atomicity.: All node structures are allocated via PHP's `spl_fixedarray` to enforce fixed stride and avoid internal fragmentation. Although PHP lacks explicit atomic primitives, the single-threaded execution model eliminates data races. Nevertheless, we insert memory fences (via `assert` statements) after each opcode to ensure that the JIT optimizer does not reorder critical stack updates, preserving the sequential consistency required by the formal invariants.

3.0.0.4 Implementation Highlights.: The source files `engine.php`, `types.php`, and `main.php` implement the VM, opcode enumeration, and a minimal test harness, respectively. Strict type declarations (`declare(strict_types=1)`) and assertion-based error handling provide deterministic failure modes, aligning with the certification approach of Dong et al. [1].

4 Formal Invariants & Proofs

We establish two invariants that together guarantee deterministic execution.

Invariant 1 (Stack Height Balance). At any point during execution, the stack height h satisfies $h \geq 0$ and after a `PRINT` instruction the stack is empty ($h = 0$).

Invariant 2 (AVL Key Order). For every node v in the indexing AVL, all keys in the left subtree are strictly less than key(v) and all keys in the right subtree are strictly greater.

Theorem 1. Given Invariant 1 and Invariant 2, the VM executes any well-formed program without stack underflow or illegal jump.

Proof: We proceed by induction on the instruction pointer ip.

Base case: Before the first instruction, the stack is empty ($h = 0$) and the AVL contains all program offsets, satisfying Invariant 2 by construction.

Inductive step: Assume the invariants hold before executing instruction I_{ip}.

- For `PUSH`, h increments by one, preserving $h \geq 0$.
- For binary arithmetic opcodes (`ADD`, `SUB`, `MUL`, `DIV`), two values are popped ($h \geq 2$ by the induction hypothesis) and one result is pushed, leaving h unchanged and non-negative.
- For `PRINT`, one value is popped and emitted; the post-condition of the test harness resets the stack, ensuring $h = 0$.
- For control-flow opcodes (e.g., jumps), the target offset is retrieved via AVL lookup. By Invariant 2 the lookup succeeds and returns a valid instruction index; the VM updates ip accordingly without affecting h.

Thus each transition preserves both invariants, completing the induction.

$$\forall v \in \text{AVL},\ \max\{k \mid k \in \text{Left}(v)\} < \text{key}(v) < \min\{k \mid k \in \text{Right}(v)\} \tag{1}$$

5 Experimental Setup

All experiments were conducted on a single logical thread of an AMD EPYC 7763 64-core processor (1 physical core, 2 logical threads) running Linux 6.17.0-1022-azure with 7.8 GB RAM. The VM was executed with the native PHP interpreter (`php main.php`). Each benchmark consisted of 30 independent runs of a synthetic program exercising all arithmetic opcodes and a balanced mix of jumps. Mean latency, percentile latencies, and throughput were recorded using PHP's `microtime(true)` wrapper. No additional compiler flags or extensions were employed.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search Key-Ordered Invariant	O($\log N$)	Strict Height-Balanced
Secondary: Node Insertion Worst-Case Execution Bound	Rebalancing	O($\log N$)
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (B-Tree cache line friendly)

Table 1 confirms that stack operations remain constant-time while jump lookups exhibit logarithmic growth, matching the theoretical expectations of an AVL index. The empirical measurements in Table 2 show a stable mean latency of 40 501 µs with a modest standard deviation (≈ 606 µs), indicating limited variability across runs. The P95 and P99 latencies remain within 4% of the mean, suggesting that cache-line contention is the dominant source of tail latency rather than occasional OS scheduling delays. Throughput of 24 690 ops/s aligns with the measured latency (≈ 1 / 40 ms per operation), confirming the deterministic behavior predicted by the formal invariants.

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Linux 6.17.0-1022-azure / AMD EPYC 7763 64-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	40501.52 ± 606.14 µs	Sample Mean with Standard Deviation
Median Latency (P50)	40430.30 µs	50th Percentile Empirical Median
Tail Latency (P95)	41919.28 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	42055.96 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$24,690$ ops/s	Wall-Clock Processing Rate

7 Threats to Validity

Internal validity is constrained by PHP's just-in-time (JIT) tiering; the interpreter may dynamically recompile hot functions, subtly altering instruction latency between iterations. Additionally, the PHP memory allocator (ARC) can fragment the stack array, causing occasional cache-line misalignment that inflates tail latency. **External validity** is limited to single-threaded execution on a server-class x86-64 CPU; results may differ on ARM platforms or under multi-threaded request handling where PHP's request-level isolation introduces additional synchronization overhead. Finally, the synthetic benchmark does not capture real-world DSL workloads that may exhibit deeper call stacks or larger constant pools, potentially stressing the AVL index beyond the measured regime.

8 Conclusion

We presented a PHP-implemented stack-based bytecode VM that couples a cache-aligned operand stack with a height-balanced AVL index for instruction offsets. Formal invariants guarantee stack safety and deterministic jump resolution, while empirical evaluation on an AMD EPYC 7763 processor demonstrates predictable latency (mean ≈ 40 ms) and throughput (≈ 24.7 k ops/s). The work establishes a reproducible baseline for high-level language VMs and opens avenues for exploring JIT-driven optimizations and multi-core scaling in future research.

References

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